Mariem Ouertani *ICT & Software Engineering Student*

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EDUCATION

Computer Networks and Telecommunication Engineering, INSAT 09/2025 – Present | Tunis, Tunisia
Top of the class, ranked 1st out of 88 — Very Good

International Summer School Program, Blockchain, University of Zurich (UZH) 07/2026 – 07/2026 | Zurich, Switzerland
- Graduated with the highest distinction (6/6), Certificate of Completion earned on an academic excellence scholarship
- Ranked 4th out of 13 international teams in the program's final evaluation; awarded a medal for the design and development of the final project, carried out within an international, multidisciplinary team

Integrated Preparatory Cycle, National entrance to engineering schools, INSAT 09/2023 – 05/2025 | Tunis, Tunisia
With Honors,
Ranked 3rd out of 86 (Top of Class)

PROFESSIONAL EXPERIENCE

Data Scientist, AI/ML Engineer (Internship), SaHera 06/2026 – Present | Montreal, Quebec, Canada
- **Literature review**: Conducted an in-depth scientific study to establish solid methodological foundations.
- **Data management**: Collected, structured, and organized a multimodal dataset.
- **Exploratory Data Analysis (EDA) & Visualization**: Performed statistical analyses and visualizations.
- **Processing pipeline**: Designed and implemented a rigorous data cleaning, preprocessing, and optimization workflow.
- **Modeling & Optimization**: Designed, trained, tuned hyperparameters, and evaluated multiple Machine Learning and Deep Learning model architectures.

RPA Developer, SAP Consultant (Internship), Iliade Consulting 07/2025 – 08/2025 | Les Berges du Lac, Tunisie
•Designed and developed an invoice automation robot in SAP IS-U using UiPath.
•Applied artificial intelligence and computer vision techniques to optimize interactions with the SAP interface.
•Project analysis and management.

PROJECTS

Adversarial Neural Cryptography, Personal Research Project (PPP) 01/2026 – 06/2026
- Implemented the **"Alice-Bob-Eve" architecture** based on the research paper "Learning to Protect Communications with Adversarial Neural Cryptography".
- Developed an end-to-end encryption system using **GANs** to learn robust protocols without classical algorithms.
- **Stack:** Python, PyTorch, NumPy, Scikit-learn.

UKF-Based CubeSat Attitude Estimation, Academic project 04/2026 – 05/2026
- Expertise: Designed a nonlinear state estimator for nano-satellite attitude tracking (Euler kinematics, magnetometer and sun-sensor fusion) with adaptive noise learning and chi-square anomaly detection.
- Key Achievement: Implemented an Unscented Kalman Filter in MATLAB that kept pointing error at 1.3° where a classic EKF diverged to 23.4° (20-run Monte-Carlo), for only 1.4× the computation time; detected 5/5 injected sensor anomalies.
- Stack: MATLAB, Unscented Transform, EKF, Mehra Adaptive Filtering (Q/R), Monte-Carlo Simulation.

CVIntelligence | Multi-Agent Candidate Analysis System, Personnal Project 04/2026 – 06/2026
- Developed a pipeline of 5 autonomous agents (Extractor, Matcher, Ranker) to automate end-to-end recruitment workflows.
- **Innovation:** Integrated a **Human-in-the-Loop** (HITL) bottleneck for human validation and leveraged **Gemini 2.5 Flash** for deep semantic scoring.
- **Deliverables:** Automated generation of professional PDF reports and a modern interactive **Dashboard (Chart.js)** for recruitment KPIs.
- **Stack:** Python, Google GenAI, PyMuPDF, ReportLab, HTML/Tailwind CSS.

SemanticResearcher | Autonomous Research & Semantic Synthesis Agent, Personnal Project 04/2026 – 06/2026
- Designed an autonomous agentic system capable of transforming vague queries into structured 8-section research reports.
- **Architecture:** Implemented the **ReAct** (Reasoning & Acting) pattern and a **Critic Agent** for self-correction (automatic retry loops based on quality scores).
- **Performance:** Drastically reduced hallucinations by cross-referencing real-time web sources (DuckDuckGo API) and performing semantic entity extraction.
- **Stack:** Python, LLM (Gemini), Agentic Design Patterns (NVIDIA DLI).

CardioSight-AI | Dual Medical AI Platform (Classification & Biometrics), Personal Project 04/2026 – 06/2026
- Developed an ECG analysis system diagnosing cardiac pathologies and identifying individuals via their electrophysiological signature.
- **Diagnosis**: ResNet1D architecture optimized for arrhythmia type detection.
- **Cardiac biometrics**: Designed "BiometricNet" for ECG-based biometric identification.
- **Stack**: Python, Deep Learning, LLM.

Smart Agri-Sensor: IoT & Edge AI Solution , Academic project 11/2025 – 12/2025
- **Expertise:** Designed an end-to-end IoT solution for precision irrigation and environmental monitoring.
- **Key Achievement:** Implemented **Edge AI** (Rule-Based Logic) on an ESP32 microcontroller for autonomous crop status classification.
- **Stack:** C++, MQTT Protocol (EMQX), Node-RED Dashboard, and low-power optimization (Deep Sleep).

REASERCH & IMPLEMENTATION

Next-Gen Networking Simulation (6G & Wi-Fi 7)
Python simulation of **OFDMA**, MU-MIMO, and **Federated Learning** for channel quality prediction in 6G networks.

Deep Learning & Transformers
Re-implementation of the **"Attention Is All You Need"** paper (Transformers) for automatic modulation classification in 5G.

AI Agents & LLM
Experiments on AI agent architecture and attention mechanisms for system optimization.

SKILLS

Languages: Python (Expert), C/C++, Java, SpringBoot, JavaScript, PHP. | **Networking:** 5G/6G, Wi-Fi 7, TCP/IP, CCNA, Wireshark. |
AI & Data: PyTorch, TensorFlow, Scikit-learn, GANs, LLMs, Computer Vision, AI Agents. | **Tools:** Git, Docker, Linux (Ubuntu/Debian), SAP, UiPath. | **Full Stack Development** |
Communication, Teamwork, Time Management, Agile Scrum, Organization, Critical Thinking, Problem Solving

CERTIFICATES

IT & AI — 50+ credentials from NVIDIA, IBM, Google, AWS, and Cisco (CCNA, Cybersecurity, Cloud). Full list and verification links available on my LinkedIn profile.

LANGUAGES

Arab — Native • **French** — Bilingual • **English** — Fluent , C1 Advanced • **German** — B1

ORGANISATIONS

Junior Entreprise , Member in Project Pole (Web Development Cell) and Commercial Development team 09/2024 – Present | Tunis, Tunisia
.Active contribution to organizing large-scale events, including the Junior Enterprise Forum 2024 and 2025, JCR 1.0, and Seneca Hack.

IEEE INSAT SB , Active Member 09/2023 – Present | Tunis, Tunisia
.Managed SPA'x events and served as Tomorrow Land Ambassador.

IEEE Computer Society Chapter INSAT SB, Member in Project Development Department 01/2025 – 05/2025 | Tunis, Tunisia
.Actively contributed to the realization of the INSAT Platform project.

AWARDS

Best Project, University of Zurich 07/2026
•Ranked 4th out of 13 international teams in the program's final evaluation; awarded a medal for the design and development of the final project, carried out within an international, multidisciplinary team